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"I believe it will not get worse": A mixed-methods longitudinal study about patient's perspective of recently developed patellofemoral pain

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ABSTRACT

Objectives: This study aimed to: (i) understand how women perceived their recently developed patellofemoral pain (PFP) regarding its cause, prognosis, and willingness to seek treatment; (ii) investigate self-reported function, knee-related quality of life (QoL), fear of movement, and physical activity level at the onset of PFP.
Design: Mixed-methods longitudinal study.
Setting: University.
Participants: Sixty-eight pain-free women were followed up over one year.
Main outcome measures: Those who developed PFP were interviewed within one month of the development of symptoms. Self-reported function, kinesiophobia, knee-related QoL, and physical activity were obtained at baseline and follow-up assessments.
Results: Twenty-one women developed PFP. Most participants reported believing the increase in physical activity and/or sitting time was associated with the onset of PFP. Many reported believing symptoms would improve over time without any treatment. Only a small number of participants intended to seek care. Quantitatively, decreases in self-reported function and QoL, as well as increases in the physical activity level were observed after PFP development.
Conclusion: Although decreases in self-reported function and QoL were observed, women reported believing their PFP is self-limiting and do not need treatment. Strategies to accurately disseminate knowledge about PFP are needed to help stimulating early care.

1. Introduction

Patellofemoral pain (PFP) is a common knee disorder that mostly affects young women (Boling et al., 2010), with an estimated prevalence of 22.7% in the general population (Smith et al., 2018). Symptoms are defined as diffuse anterior knee pain aggravated by activities that overload the knee joint (e.g., stepping tasks, squatting, running, landings) (Crossley et al., 2016). People with PFP typically report reduced physical activity levels (Glaviano et al., 2017, 2022), functional capacity (Waiteman et al., 2024), and quality of life (QoL) (Djurtoft et al., 2023). Higher pain catastrophizing and fear avoidance beliefs are also reported (Vicenzino et al., 2022) and associated with higher levels of pain and disability (Botta et al., 2023). The possible progression to patellofemoral osteoarthritis further highlights the importance of understanding the development of PFP and its management (Crossley, 2014; Utting et al., 2005).

The exact reason why people develop PFP is not completely elucidated (Crossley et al., 2016). Sudden increase in physical activity levels (i.e., doing too much, too soon) seems to be an important trigger (Crossley et al., 2019; Dye, 2005; Willy et al., 2019). Quadriceps weakness, but not multiple variables (including sex, body mass index, and Q-angle), has also been reported as a risk factor for future PFP (Neal et al., 2019). In terms of PFP management, current evidence-based "best management" available is exercise therapy and patient education (Collins et al., 2018; Willy et al., 2019). However, the prognosis of PFP is still unfavorable for up to 60% of the patients, who report symptom

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recurrence 5–8 years after treatment (Lankhorst et al., 2016).

The existing evidence regarding PFP risk factors, management, and prognosis was summarized in a clinical practice guideline published in 2019 (Willy et al., 2019) and later disseminated to the public via infographics, podcasts (Ardern et al., 2020; *JOSPTasks*, 2020), patient (Teyhen & Robertson, 2018, 2019) and clinician perspectives (Scott et al., 2019). Despite this, recent surveys have reported that physical therapists and athletic trainers still have incorrect beliefs regarding risk factors for PFP (e.g., hip strength weakness as a risk factor) (Willy et al., 2022; Zambarano et al., 2022). Incorrect beliefs in terms of the benefits of passive treatments for PFP and that PFP does not result in unfavorable long-term outcomes were also identified (Willy et al., 2022; Zambarano et al., 2022). These findings outline that, despite the existing evidence, inaccurate beliefs towards PFP are common.

Although recent studies have investigated the knowledge of health professionals in providing evidence-based care to people with PFP, no study has sought to determine how the patients themselves perceive this musculoskeletal condition at the onset of symptoms. It is unclear what patients think about the causes of their symptoms, how they will evolve, and how they intent to proceed in terms of management. Investigating patient's beliefs at the onset of symptoms can improve our understanding about how debilitating they think PFP may be, which can affect their willingness to seek medical care for it.

Therefore, the aim of this study was to understand and explore how women perceived their first PFP experience regarding its cause, prognosis, and willingness to seek treatment. The secondary aim of this study was to investigate self-reported function, knee-related QoL, kinesiophobia, and physical activity level at the onset of knee pain.

2. Materials and methods

2.1. Study design and participants

This was a convergent mixed method longitudinal study. Both quantitative and qualitative analyses were utilized to deliver a comprehensive evaluation of participants' clinical status, as well as their subjective perception, expectations, and willingness to seek treatment. Reporting of this mixed-methods study was performed according to the Good Reporting of A Mixed Methods (GRAMMS) Study Checklist (O'Cathain et al., 2008), and Standards for Reporting Qualitative Research (SRQR) (O'Brien et al., 2014). The principles and methods of phenomenology guided the qualitative data collection and analysis, which seeks to describe the meaning and significance of human experiences (Merriam, 2009). This study was approved by the local ethics committee (number: 4.649.629) and all participants provided written informed consent to participate. The present study complies with the Declaration of Helsinki.

Sixty-eight pain-free women were recruited via advertisements and flyers distribution at universities, gyms, and posts on social media. Purposeful sampling was used to ensure the inclusion of participants according to the following criteria: i) age between 18 and 25 years; ii) no current signs or symptoms or history of any musculoskeletal disease or trauma to the knee in the last six months; iii) no self-reported pain coming from the spinal column, history of previous surgery, or neurological disease; iv) no physical therapy (at least six months) or drug treatment (prescribed nonsteroidal anti-inflammatory drugs or pain killers) for any lower limb pain and/or injury. Recruitment continued until data saturation during follow-up, determined by two consecutive interviews not identifying new themes. Although the concept of data saturation has become contentious in qualitative research (Virginia Braun & Clarke, 2021), the decision to recruit until data saturation aligns with the SRQR checklist compliance (O'Brien et al., 2014).

2.2. Procedures

Baseline assessments were performed at the Sao Paulo State University (Presidente Prudente, Sao Paulo, Brazil). Demographics collected included age, body mass, height, and educational level. All participants also completed the Anterior Knee Pain Scale (AKPS), the Tampa Scale for Kinesiophobia (TSK), the Knee Osteoarthritis Outcome Score (KOOS), the International Physical Activity Questionnaire Scoring Protocol short form (IPAQ-S) and rated their worst knee pain intensity in the last week using a 100 mm visual analogue scale (VAS). These tools are valid and commonly used to assess individuals with PFP (Craig et al., 2003; Crossley et al., 2004; Cunha et al., 2013; De Souza et al., 2008; French et al., 2007; Gonçalves et al., 2009).

The AKPS is a 13-item questionnaire that assesses knee-related self-reported function, with the scores ranging from 0 (maximal disability) to 100 (no disability) (Crossley et al., 2004; Cunha et al., 2013). The TSK is a 17-item questionnaire that assesses pain-related fear associated with movements and physical activity, with the scores ranging from 17 (lower fear) to 68 (higher fear) (De Souza et al., 2008; French et al., 2007). The KOOS is a 42-items questionnaire that assess health problems related to the knee through subscales (symptoms, pain, activities of daily living, sports, and knee-related QoL); with the scores ranging from 0 (extreme problems) to 100 (no problems) (Gonçalves et al., 2009). For the purpose of assessing knee-related QoL, only the QoL subscale from KOOS was considered for analyses. The IPAQ-S is a questionnaire that assesses self-reported physical activity in METs (MET-min/week) (Craig et al., 2003). The total score was calculated for the previous seven days using the sum of: walking (3.3 METs*min*days), moderate intensity activities (4.0 METs*min*days), and vigorous intensity activities (8.0 METs*min*days) (Craig et al., 2003). The VAS consists of a 0–100 mm horizontal line, with 0 representing "no pain" and 100 representing "extreme pain" (Crossley et al., 2004).

After baseline assessments, participants were followed up over a one-year period. They were oriented to contact a researcher by telephone or email (MW) if they started having anterior knee pain. Those who developed insidious pain behind or around the patella were identically reassessed within one month of the appearance of an insidious onset knee pain (Follow-up 1 – Pain evaluation). Knee pain had to correspond to at least 30 mm on the VAS, aggravated by at least one activity that overloaded the patellofemoral joint (e.g., stair negotiation, running, jumping), as recommendations of the latest consensus statement on clinical examination of PFP (Crossley et al., 2016). The outcomes reassessed were self-reported function, knee-related QoL, fear of movement, and physical activity level. Subsequent one-on-one semi-structured in-person interviews were performed by one female researcher (MW), with more than five years of experience assessing individuals with PFP. Before starting to record the interviews, the researcher introduced herself as physiotherapist and researcher conducting a master's degree and explained the interest to know more about patients' perspectives about their recently developed pain. During the interview only the participant and the researcher remained in the room. No assumptions, reasons, and personal interests from the researcher were reported to the participants.

In order to standardize the interviews, a pilot study was performed where the researcher interviewed five women for training and experience. The interviewer did not have any existing relationship with participants and the same questions were applied to all participants. A topic guide was used to conduct the interview process and developed by the authors: (1) Why do you think your knee has started to hurt? (2) How do you think your condition will evolve? (3) Do you intend to do something to reduce your pain? In case participants did not fully understand the question, the interviewer reformulated the same question in a different format, allowing a flexible and more comprehensible interview. All interviews were audio-recorded and no field notes or repeat interviews were performed.

2.3. Data analysis

Quantitative data of self-reported function, kinesiophobia, knee-related QoL, and physical activity were exported from REDCap® to

SPSS version 23.0 (IBM Corp. Released 2015. IBM SPSS Statistics for Windows, Armonk, NY: IBM Corp.) for statistical analyses. Data from baseline and follow-up assessments were then compared using dependent t-tests ($p < 0.05$). Effect sizes (Cohen's *d*) were also calculated and interpreted as 'small' (0.20–0.49), 'medium' (0.50–0.79), 'large' (0.80–1.29) and 'very large' (≥1.30) (Cohen, 1988).

Two physical therapists with experience assessing patients with PFP led the qualitative data analysis. Qualitative data analysis was performed using a six-step inductive thematic analysis: transcription, familiarization with data, coding, developing an analytical framework, applying the framework, and charting data to the framework matrix (Braun & Clarke, 2006; Gale et al., 2013). One researcher (LP) transcribed the interviews verbatim and a second researcher (LF) double-checked the transcriptions with the audio-recorded interviews for accuracy. Transcripts were independently coded.

Audio files and transcripts were listened/read several times by both researchers (LP and LF) to allow 'data immersion' and gain familiarity. Condensed meaning units were abstracted, coded, and then used to divide information into similar categories. Data categories were independently developed to identify themes and subthemes (Braun & Clarke, 2006; Charmaz, 2006; Gale et al., 2013). Four meetings were held between the two researchers to refine the thematic framework, and themes and subthemes were finalized through discussion. Any disagreements were discussed, and themes and subthemes were edited until consensus was reached. After definition of themes and subthemes, the transcripts were read again to ensure that data were not missed or misinterpreted. Quotes were then organized and combined to fit each theme and subthemes based on data that were transcribed and coded. One researcher (LP) identified supportive quotes and a second one (LF) double checked them for agreement.

2.4. Reflexivity and trustworthiness

Reflexivity was carried out through keeping a shared record of the researchers' preconceptions around the research area. Researchers reflected on their thoughts, feelings, and decision-making process for theme and subtheme generation and their potential impact on data analysis. To enhance trustworthiness, peer debriefing was conducted to ensure systematic analysis and verify that each theme and subtheme had a coherent pattern (Nowell et al., 2017). An audit trail of the analytic process was kept, and relevant, rich quotations from participants' accounts were provided.

3. Results

Of the sixty-eight participants, twenty-three women reported PFP during the follow-up period (mean 99.33 ± 61.90 days after baseline assessment) with two of them being excluded from the analyses due to the report of trauma on their knees between the baseline and follow-up assessments (Fig. 1). Twenty-one women who reported insidious anterior knee pain were interviewed. Demographic data from participants who developed PFP are presented in Table 1.

3.1. Qualitative data

Three themes emerged from the analyses: i) Change of habits; ii) Beliefs about prognosis and iii) Measures to reduce the knee pain. Each of these were relevant to the original research question. Each theme and respective subthemes are discussed below, including supporting quotes from the participants and their identifying number.

3.1.1. Theme 1: change of habits

Approximately 86% (n = 18/21) of the participants reported believing their pain emerged due to changes in their daily living habits, such as increases in sitting/standing time, engagement into sports practice, and physical activity modifications (Table 2). The most reported changes in daily living activities were the increase in sitting or standing time (n = 10), followed by sudden increases in walking or running practice (n = 6), and increase in general physical activity level (n = 2). Three participants reported no changes in their habits and could not identify a possible reason/cause for their knee pain.

3.1.2. Theme 2: beliefs about prognosis

Fifty-seven percent of the participants (n = 12/21) reported believing that their pain is self-limiting and would improve or get better over time with no need for any specific treatment (Table 3). However, 71.4% of them (n = 15/21) reported they intended to create strategies to prevent further symptoms. Some of the strategies included remaining as active as possible and performing load management by themselves. Participants who were less worried about their knee pain reported expecting to improve over time as they get used to the modifications on physical activity level (i.e., they reported believing pain is a normal process of starting or increasing physical activity).

Two participants expected to get worse over time, however both seemed willing to start or continue exercising. Seven participants reported believing their symptoms would persist over time. Despite these beliefs, most participants reported thinking that pain would remain the same (i.e., symptoms will not get better or worse), and they would not

Table 1
Demographic characteristics of the participants who developed patellofemoral pain.

Variables	Baseline data mean (standard deviation) N = 21
Age (y)	19.19 (1.88)
Body mass (Kg)	59.62 (10.79)
Height (m)	1.63 (0.06)
Body mass index (Kg/m^2)	22.39 (4.00)

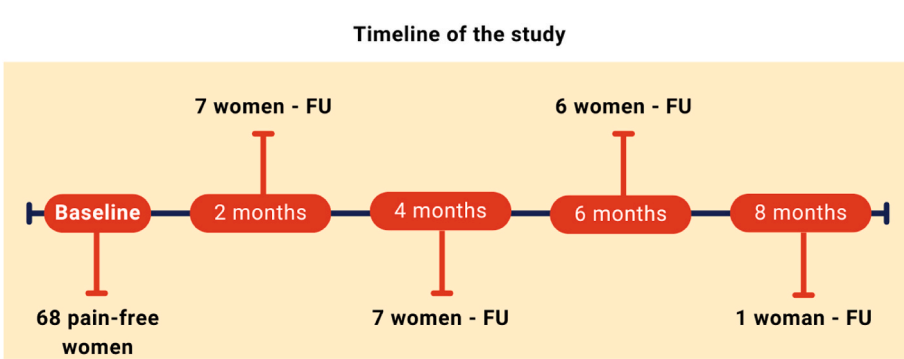

Fig. 1. Timeline representing the frequency of PFP reports over the study period.
'FU' indicates follow-up or "pain assessment".

Table 2
Illustrative quotes about the first theme - Change of habits.

Subthemes	Illustrative quotation
Prolonged sitting or standing time	● P1: "I think I've never paid attention (at my knee). I think it's because I spend lots of time sited at college and at home." ● P3: "maybe it's because I remained seated for a long time." ● P5: "I think because I've been standing for a long period after starting a new job." ● P15: "I used to remain seated for a long time. Then I started feeling it."
Sports practice engagement	● P4: "I think it's because I started exercising more often and my shoes were old." ● P14: "I think it was because I wasn't very used to exercise and I did a 10 km walk." ● P16: "Because I wasn't used to run, and I started to run a lot." ● P20: "I didn't exercise at all. I just walked a lot, and it started hurting."
Physical activity level modification	● P6: "It could be because of the effort made to walk more." ● P9: "Because I was doing more exercise than I used to." ● P21: "I started to do more physical activity."
Unspecified reasons	● P11: "I have no clue. I haven't changed anything in my routine." ● P12: "I don't know. I haven't changed anything, I haven't done anything, I believe it is something normal." ● P13: "I don't know. I haven't changed anything in my routine."

Table 3
Illustrative quotes about the second theme - Beliefs about prognosis.

Subthemes	Illustrative quotation
Symptoms would get better	● P2: "I think it's an isolated case and it won't hurt anymore." ● P4: "I think it will stop soon." ● P8: "I hope it doesn't hurt anymore. I have been noticing that it isn't hurting that much." ● P11: "I think I won't feel any more pain." ● P13: "I hope it doesn't hurt anymore." ● P16: "I think it's ok because I'm getting used to running. I think that as soon as I get used to it, I won't feel it (pain) anymore." ● P17: "I think it (the pain) will get better." ● P20: "Oh, I hope it (the pain) gets better."
Symptoms would get worse	● P1: "I intend to prevent this (the pain recurrence). I hope it doesn't get worse, but if I feel I may probably get worse if I remain seated for a prolonged time without doing any exercise." ● P9: "I don't know, I think it might hurt a little more because next year I intend to come back to the gym."
Symptoms would remain the same	● P3: "I may have some discomfort from time to time, but I think it won't bother me for so long" ● P5: "It won't get worse. I think I may feel it when I push a bit harder." ● P7: "I think it will continue as it is. I hope it doesn't hurt anymore and not progress to something worse." ● P10: "I think it might hurt again sometimes, but I don't think it's going to be constant." ● P18: "I believe it will not get worse." ● P21: "I think it won't get much worse. It will stay as it is now."
Do not know	● P6: "I don't know."

have severe daily life limitations in the future.

3.1.3. Theme 3: measures to reduce the knee pain

A small number of participants intended to seek medical care (n = 5). Three of them reported they would seek care only if the symptoms got worse (i.e., if pain becomes more frequent or starts to interfere in daily-life activities) (Table 4). Participants who reported being worried about their knee pain also reported they would seek a medical doctor as the first contact healthcare professional because they felt the necessity of having imaging exams to better understand the causes of their symptoms. Participants who reported being less worried about their pain reported self-administration (i.e., self-care) of pain reliefs strategies (e.g., use of ice bags or non-prescribed pain killers).

Most of the participants (71.4%) seemed to understand that exercise is essential to prevent poor prognosis. In fact, they were willing to remain active or make physical activity modifications to improve their pain. Only one participant reported the thought of stopping exercising to improve pain and 3 participants reported they would do nothing to manage their pain.

3.2. Quantitative data

Statistically significant decreases in self-reported function and knee-related QoL, as well as an increase in the physical activity level were reported after the development of PFP (Table 5). No statistically significant changes in kinesiophobia were reported.

4. Discussion

In our study, women with recently developed PFP associated the onset of their symptoms with an increase in sitting or standing time, as well as a sudden increase in walking, running, and/or general physical activity. Most participants reported believing that their pain is self-limiting and it would not worsen over time, or even improve as its natural course. This belief may elucidate why only a small number of participants intended to seek medical care. Quantitative data showed worse self-reported function and knee-related QoL, as well as an increase in physical activity level. Our findings highlight the importance of patient education about PFP during first consultation given the inaccurate beliefs about PFP and its clinical course. Knowledge dissemination about PFP should also be encouraged to the general adult population, especially those who are at greater risk of developing PFP (e.g., runners,

Table 4
Illustrative quotes about the third theme - Measures to reduce the knee pain.

Subthemes	Illustrative quotation
Seek medical care	● P1: "I intend to go to the doctor first to have imaging exams." ● P2: "I intend to go to the doctor if necessary. I also intend to practice more exercise." ● P4: "I don't know. I may stop exercising first. Then, if its (the pain) don't get better, I maybe go to see the doctor." ● P9: "If it (the pain) gets too often, I may look for some help."
Self-care	● P6: "I take pain killers and use cold-water bag when it (the pain) comes." ● P8: "I intend to take pain killers." ● P11: "I may take some pain killers."
Continue exercising or perform activity modifications	● P1: "After seeking medical care I intend to start doing some exercise, something to prevent it (pain recurrence) and see if it doesn't evolve." ● P3: "I consider start doing some exercises next year." ● P5: "I just want to come back to my regular work out." ● P18: "I intend to do more exercise to reduce my sitting time." ● P19: "I've been stretching in the morning to see if it helps me to get better." ● P21: "I think I should continue doing some exercise to see if it goes away."
Do nothing	● P12: "No, because it doesn't bother me." ● P13: "I don't think so, it is not a big deal." ● P17: "As it was not intense, I don't think so."

Table 5
Quantitative data of participants who developed PFP at baseline (pain-free) and follow-up (pain) assessments.

Patient-reported outcomes	Baseline: n = 21	Follow-up n = 21	Mean Difference (95% CI)	p-value	Effect size^{Interpretation}
Function (AKPS)	96.00 (4.80)	91.43 (5.99)	5.50 (1.92; 9.08)	0.004[a]	0.81^{Large}
Kinesiophobia (TSK)	35.50 (5.47)	33.00 (6.09)	2.30 (−1.38; 5.98)	0.214	0.43^{Small}
Knee-related QoL (KOOS)	99.10 (2.93)	92.71 (8.70)	7.05 (2.84; 11.34)	0.002[a]	0.98^{Large}
Physical-activity (IPAQ-S)	1086.32 (933.18)	2528.85 (2814.94)	−1442,52 (−2813.89; −71.15)	0.040[a]	0.68^{Medium}

[a] Represents statistical significance ($p < 0.05$).

military).

Intense and frequent sports participation, or sudden increases in physical activity, with insufficient recovery or inadequate load management are thought to contribute to the development of PFP (Hall et al., 2015). Findings from our study indicate this is also a common belief among women who have recently developed PFP. Self-reported physical activity obtained with IPAQ-S questionnaire may support this notion given the statistically significant increase in physical activity levels. It could be that their knee joint was exposed to repetitive and/or higher loading through high volume of physical activity without having any gradually exposure or preparation (Hall et al., 2015). Treatment strategies focused on activity modification and patient education on load management have been associated with high rates of successful outcomes among adolescents and runners with PFP (Esculier et al., 2018; Rathleff et al., 2019). However, to our knowledge, no study has investigated their effects on preventing PFP.

Some participants also associated the development of their symptoms with increases in prolonged sitting or standing time. Pain during prolonged sitting is commonly reported by individuals with PFP (Collins et al., 2016; Glaviano, Holden, et al., 2022). A recent systematic review has reported that sitting resulted in greater pain severity when compared to squatting, running, walking, knee extension, patellofemoral joint loading, lunge, and stair ambulation tasks (Glaviano, Holden, et al., 2022). It has been suggested that the amount of knee flexion while sitting and the prolonged nature of the task may play an important role in pain (Glaviano, Holden, et al., 2022). However, the literature regarding the association between PFP and prolonged sitting is scarce and future studies are warranted.

Most of the participants reported believing that PFP is self-limiting and symptoms would fade away as part of its natural course. Some reported believing they had to endure symptoms for a period of time and improvements would naturally happen once they were through this initial phase. Despite this being a common belief (Rathleff, 2016), PFP has been reported to persist up to 18 years after its diagnosis (Stathopulu & Baildam, 2003). PFP is not self-limiting and long symptom duration has been reported to be a predictor of poor outcome over time (Collins et al., 2010). Early timing of physical therapy has been recently reported to decrease the odds of recurrent knee pain (Young et al., 2021a). These findings highlight the importance of patient education on PFP prognosis and early treatment, which aligns with recent evidence about the necessity of efforts to mitigate early care void following knee injuries (Holm et al., 2023).

It is likely that participants did not intend to seek medical care because of the acuity of symptoms, their beliefs that pain is self-limiting, and the relatively mild symptoms and worsening in physical function and QoL. Only a limited number of participants reported that they might seek medical care due to the necessity of having a specific diagnosis based on imaging exams, which is also an inaccurate belief (the linkage between structural abnormalities and PFP is not clear) (Wyndow et al., 2016). In fact, recent evidence indicates that two in every three patients did not seek additional care after PFP diagnosis (Young et al., 2021b). The fact that symptoms often start mild and are intermittent may make patients believe that symptoms will decline soon, and medical care might not be needed. However, early treatment reduces healthcare use, costs, and the odds of having knee pain recurrence compared to having delayed treatment (Lankhorst et al., 2016; Young et al., 2021a). This suggests the longer the PFP duration with no treatment, the greater the odds of a poor prognosis/recovery (Lankhorst et al., 2016). As such, knowledge dissemination about PFP to the general population at greater risk of developing PFP (e.g., runners, military) and health care professionals may help better aligning peoples' beliefs with current evidence and stimulating early care. Further education to better align health care professionals' (e.g., athletic trainers, physical therapists, personal trainers, etc.) knowledge with the current evidence related to PFP is also important, as they can disseminate this knowledge to the general adolescent and young adult populations that may develop PFP.

This study has limitations that should be acknowledged. The sample included only young women. Although PFP is highly prevalent in this population (Smith et al., 2018), the findings may not be generalizable to men and adolescents. Similarly, no subgroups of individuals at greater risk of developing PFP (e.g., runners, military) were specifically investigated in this study. While these subgroups might have different mindsets (e.g., greater resilience), their beliefs may not be the same. Future studies investigating these populations are warranted. Lastly, only 21 women developed PFP. Although this is a limited sample size, data saturation was used to end participant recruitment.

5. Conclusion

Findings from this study indicate that most participants reported believing the increase in physical activity and/or sitting time was associated with the onset of PFP symptoms. However, many reported believing knee pain would be self-limiting, which may explain the lack of interest in seeking medical care. Therefore, creating strategies to disseminate accurate knowledge about PFP and educate people at greater risk of developing PFP as well as health care professionals may help better aligning peoples' beliefs with current evidence and stimulating early care.

Ethical statement

The authors of this manuscript confirm that this study has been carried out in accordance with The Code of Ethics of the World Medical Association - Declaration of Helsinki. Ethical approval for this study was received by Ethics Committee of São Paulo State University (registration number 4.649.629).

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CRediT authorship contribution statement

Liliam Barbuglio Del Priore: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Ronaldo V. Briani:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Marina C. Waiteman:** Writing – review & editing, Project administration, Formal analysis. **Lucca André Liporoni Bego Farinelli:** Writing – review & editing, Methodology,

Formal analysis. **Gleison Gustavo Moraes da Silva:** Writing – review & editing, Methodology, Data curation. **Théo Muniz de Souza Borges da Silva:** Writing – review & editing, Project administration, Methodology. **Neal Glaviano:** Writing – review & editing, Supervision, Data curation. **Fábio M. de Azevedo:** Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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